AI tool for optimization of design w.r.t Class rules, standards, regulations etc. and synchronize 3D models with Bill of Material (BOM) to eliminate human errors (Sl No. 71)

Executive Summary:

Mazagon Dock Shipbuilders Limited (MDL) frequently encounters design modifications during the finalisation of detail design at the stage of detail engineering of MDL designed ships due to evolving customer requirements, IMO and class rule upgrades, regulatory updates, and integration challenges. This has considerable impact of basic drawing approval from classification society and may call for multiple revisions in the approval process. Needless to mention that the work of issue of production drawings commences from first revision of approved drawings itself.

This challenge is to develop an AI-Based tool for optimising design with respect to rules, standards, regulations etc for assistance during detail design for seeking approval from classification society.

The toll shall be able to synchronize 3D model with bill of Material (BOM) so that latest approved revision is reflected in the model from which production drawings shall be generated and issue to production with implemented changes of latest revision.

Introduction & Problem Statement:

Design changes are inevitable in shipbuilding projects, arising from evolution of customer requirements, supplier innovations, classification society feedback, or integration discoveries. Current data from commercial projects indicate that 25% of changes occur post contract design finalisation impact likely to propagate across hull, propulsion, weapons, and concerned systems.

These propagated impacts may comprise of :

- Structural re-analysis and weight redistribution
- Re-qualification testing and trials

The problem statement is to develop an AI-powered system that analyzes detail design parameters with respect to class rules, standards, regulations and predicts downstream impacts across disciplines, and provides intelligent inputs towards finalisation of detail design, incorporation of the design into the basic drawings before submission to the class and incorporation of latest approved revision in the production drawings post approval of the class.

Tasks to Perform:

1. Build AI models to highlight the changes in design drawings with respect to classification rules, integration requirements , regulatory requirements etc
2. Develop an intelligent system to incorporate the changes in detail design in 3 D model
3. Develop an intelligent system to reflect the said changes in approved drawings in production drawings